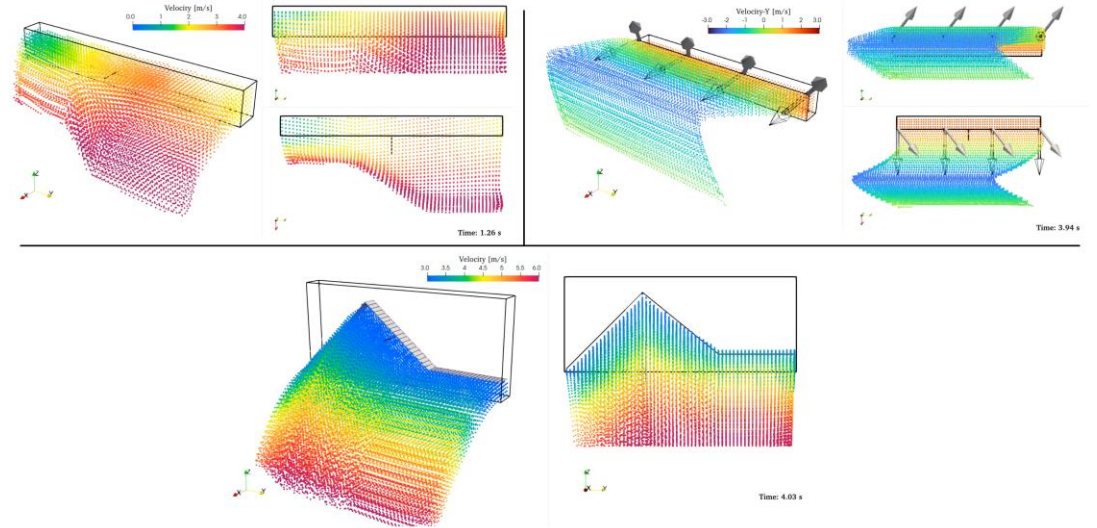


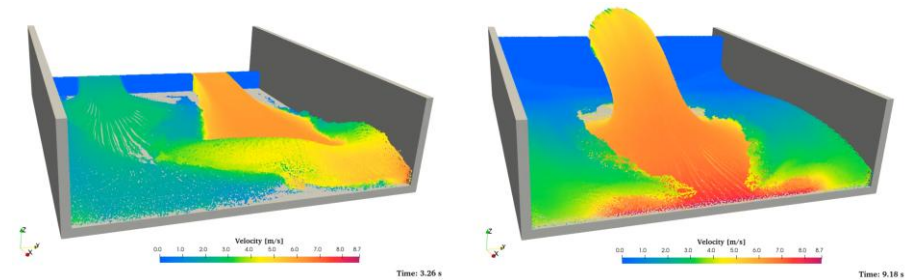
01_BASIC

- Basic example of inlet changing velocity magnitude along the inlet area during simulation according to mesh data in a CSV file. [Video](#)
- Basic example of inlet changing velocity direction along the inlet area during simulation according to mesh data in a CSV file. [Video](#)
- Basic example changing height of the inlet zone during simulation according to mesh data in a CSV file. [Video](#)



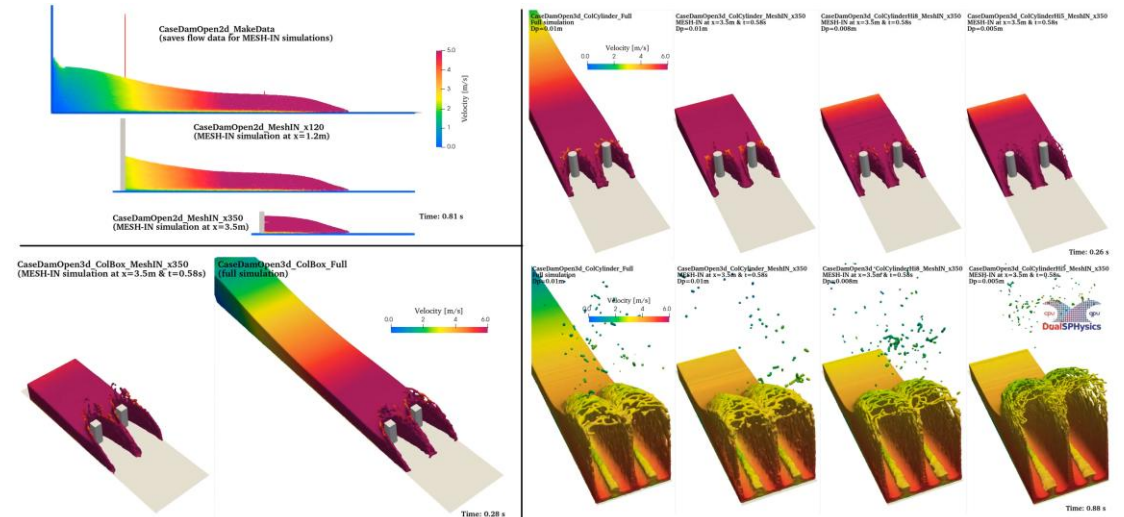
02_COMPLEX

- Complex inlet behaviour changing velocity and height during simulation according to mesh data in two CSV files generated by Python scripts in a simple way. [Video](#)



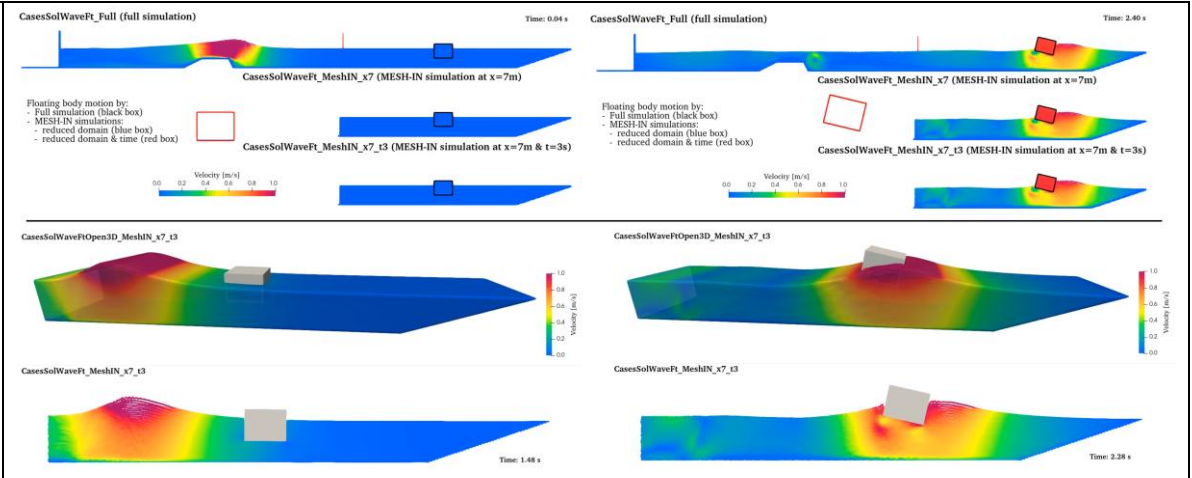
03_DAMBREAK

- 2-D application of the MESH-IN technique for dam break simulation. Flow mesh data collected from a previous simulation is used in a reduced and very reduced simulation with the same flow behaviour. [Video](#)
- 3-D MESH-IN application to simulate a dam break flow impacting on two structures to reduce the number of particles and the runtime. [Video](#)
- 3-D MESH-IN application to simulate a dam break flow impacting on two cylinders to reduce the number of particles and the runtime. [Video](#)
- 3-D MESH-IN application on dam break impacting on two cylinders to reduce the size of the simulation and allow for increased resolution in the impact area. [Video](#)



04_SOLITARYWAVEFT

- 2-D focused wave simulation with MESH-IN to reduce wave propagation time and runtime. [Video](#)
- 3-D focused wave simulation with open boundaries starting from mesh data of 2-D simulation using MESH-IN close to the floating body under study. [Video](#)



05_DAMBREAKGATE

- MESH-IN technique validation against a three-dimensional dam-break experiment with a vertical opening gate. Simulation is based on the work published by [Ruffini et al. \(2022\)](#). [Video](#)

